

## Factors influencing the structure and function of the small intestine in the weaned pig: a review

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Accepted 25 March 1997

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### Abstract

At weaning, the young pig is subjected to myriad of stressors (e.g. change in nutrition, separation from mother and littermates, new environment) which cause reduced growth. This post-weaning 'growth check' continues to represent a major source of production loss in many commercial piggeries. Associated with weaning are marked changes to the histology and biochemistry of the small intestine, such as villous atrophy and crypt hyperplasia, which cause decreased digestive and absorptive capacity and contribute to post-weaning diarrhoea. In this review we have outlined the major factors implicated in the aetiology of these changes, such as the role of enteropathogens, transient hypersensitivity to dietary antigens, and the withdrawal of milk-borne, growth-promoting factors. Special attention has been paid to the role of food (energy) intake as a mediator of intestinal structure and function after weaning, although other influences such as the source of protein added to the diet may interact with food intake to alter gut structure and function. This is clearly an area of production concern, and future research into areas such as manipulation of the immature digestive tract with exogenous growth factors and (or) dietary supplementation with 'non-essential' amino acids such as glutamine, appear warranted. © 1997 Elsevier Science B.V.

**Keywords:** Pig; Weaning; Small intestine; Villous height; Crypt depth

### 1. Introduction

There is little doubt that low voluntary food intake and associated poor growth after weaning are major limitations to enhanced efficiency in pig production. But despite the large volume of literature

dealing with the nutritional, behavioural, health and environmental needs of the weaned pig, it is apparent that the post-weaning growth check still represents a major production penalty. The marked changes that occur in gut structure and function after weaning, such as villous atrophy and crypt hyperplasia, are generally associated with the poor performance observed as they are thought to cause a temporary decrease in digestive and absorptive capacity of the small intestine. However, the precise aetiology of these changes and their relationship to production parameters in the post-weaning period are not clear. The aim of this review is to first describe the physi-

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ology and biochemistry of the small intestine in the weaned pig, and then review factors affecting its structure and function. Advances in our understanding of these fields should assist in reducing the production penalty associated with weaning.

This review is divided into several sections. A description of the histological and biochemical changes that occur in the small intestine after weaning is followed by a critical discussion of the various factors that have been implicated in causing structural and functional changes in the gut after weaning. Considerable attention is paid to the concept of luminal nutrition and its role in determining the mucosal structure and function in the weaned pig. Much of the original research into the stimulatory effects of luminal nutrition on the gut was conducted in rodents 20 to 30 years ago, perhaps begging the question of its relevance to this review. However, we would argue that including a précis of this field of research is necessary since scientists working in weaner pig research have, in general, not considered the important role food (energy) intake *per se* plays in determining gut structure and function after weaning.

## 2. Structure and function of the small intestine after weaning

The anatomy and morphology of the small intestine of the pig has been described previously in books or reviews by Nickel et al. (1973), Caspary (1987), Friedrich (1989), Kelly et al. (1992a), and Cranwell (1995), and will not be reiterated here.

### 2.1. Villous height and crypt depth

Villous atrophy after weaning is caused by either an increased rate of cell loss or a reduced rate of cell renewal. If villous shortening occurs via an increased rate of cell loss, then this is associated with increased crypt-cell production and hence increased crypt depth (e.g. microbial challenge, antigenic components of feedstuffs). However, villous atrophy might also be due to a decreased rate of cell renewal that is the result of reduced cell division in the crypts (e.g. fasting). While both these events are likely to operate after weaning to reduce the villous height: crypt depth

ratio, it is likely that the former will have the most profound effect on gut structure. The extent of cell proliferation in the crypts and enterocyte loss from the villi are modified by the type of microbial flora present and the type of diet fed, and this will be discussed later in the review.

Many authors have reported that there is a reduction in villous height (villous atrophy) and an increase in crypt depth (crypt hyperplasia) after weaning (Hornich et al., 1973; Gay et al., 1976; Kenworthy, 1976; Hall et al., 1983; Smith, 1984; Hampson, 1986a,b; Miller et al., 1986; Cera et al., 1988; Dunsford et al., 1989; Hall and Byrne, 1989; Kelly et al., 1990b, 1991b,c; Li et al., 1990, 1991a,b; Nabuurs et al., 1993a,b; Makkink et al., 1994; Beers-Schreurs et al., 1995; McCracken et al., 1995; Pluske et al., 1996b, Pluske et al., 1996c). These morphological changes are more conspicuous when weaning occurs earlier at 14 days rather than later at 28 days of age. In a comprehensive study of changes in the structure of the small intestine in the weaned pig, Hampson (1986a) reported that following weaning at 21 days of age, the villous height was reduced to around 75% of pre-weaning values within 24 hours (940 to 694  $\mu\text{m}$ ). Subsequent reductions in villous height along the small intestine were smaller but continued to decline until the fifth day after weaning, at which point the villous height at most sites along the gut was approximately 50% of initial values found at weaning (Fig. 1). Similar decreases in villous height were reported by Miller et al.

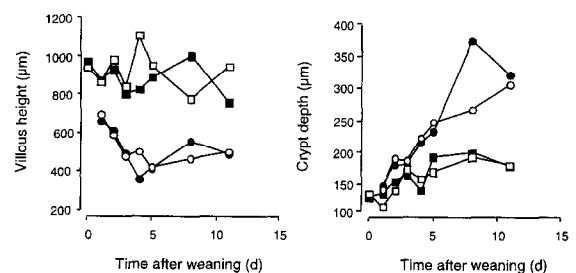


Fig. 1. Villous height and crypt depth at a site 25% along the length of the small intestine of weaned and unweaned pigs killed between 21 and 32 days of age: ○—○ pigs weaned at 21 days of age and offered creep food prior to weaning; ●—● pigs weaned at 21 days of age but not offered creep food prior to weaning; □—□ pigs unweaned and offered creep food; ■—■ pigs unweaned but not offered creep food. Values are means for between two and seven pigs killed per treatment combination per day (redrawn from Hampson, 1986a).

(1986), while Cera et al. (1988) additionally reported a reduction in the length of microvilli three to seven days after weaning. From five to eight days after weaning the villous height began to increase. In contrast, unweaned pigs showed only slight reductions in villous height. The longer villi present in the proximal part of the small intestine decreased in height proportionally more than the villi towards the distal part of the gut. Villous atrophy was caused by a reduction in the number of enterocytes lining the villus and was not due to villous contraction, a phenomenon suggested by Hampson (1986a) to represent either an increased rate of cell loss from the villous apex or a brief reduction in the rate of cell production in the crypts.

A decrease in crypt-cell production rate associated with villous atrophy was also reported by Hall and Byrne (1989), a mechanism attributed to sub-optimal intakes of energy and protein. Since crypt depth was reduced at three days after weaning, Hall and Byrne (1989) suggested that villous stunting was due to a slowed production of new cells and not an accelerated rate of loss of mature enterocytes from the surface of the villi. Hampson (1986a) reported that the number of cells in the crypts was not increased two days after weaning, but increased steadily thereafter until the eleventh day. Crypt elongation also occurred in unweaned pigs, but the increase was greater in weaned animals.

As a result of these changes in villous height and crypt depth after weaning, the villous height: crypt depth ratio in weaned pigs is markedly reduced compared to unweaned animals. Hampson (1986a) suggested that this represented a balance of cell production in the crypts and cell loss from the villi that began on the fifth day after weaning and persisted for at least five weeks. This is manifested in a change from the longer, finger-like villi seen in newborn and sucking pigs to wider leaf-like or tongue-like villi.

These changes only occur after weaning if there is continuous absence from the sow, since pigs weaned for two days and then returned to the dam for three days showed crypt elongation only equivalent to that of pigs weaned for two days (Hampson, 1983). These changes also occur at a time when growth per se of the small intestine is extremely rapid. Goodlad and Wright (1990) and Kelly (1994) suggested that

this can be explained by the young animal devoting a considerable part of its intestinal cell differentiation to cryptogenesis rather than the influx of new cells onto the villi, an event likely to be under a degree of genetic control. Furthermore, it appears that consumption of food after weaning is necessary for crypt hyperplasia to occur, but a lack of intake may not be necessary for villous atrophy to occur (Hampson, 1983).

## *2.2. Digestive and absorptive capacity of the small intestine after weaning*

Reduction in villous height and increases in crypt depth in the small intestine after weaning are generally associated with reductions in the specific activity of the brush-border enzymes lactase and sucrase (see Pluske et al., 1995), although the large variation associated with assay techniques and the variability between studies has not always resulted in a statistical decline in activity.

Hampson and Kidder (1986) reported large, rapid reductions in the specific activity of lactase and sucrase that reached minimum values four to five days after weaning and occurred regardless of whether creep food was offered prior to weaning or not. Whereas lactase continued to decline at all but the most distal sites along the gut, sucrase activity had recovered by 11 days after weaning. The greater loss in lactase than sucrase activity was most likely due to the more apical distribution of lactase activity along the villus (Tsuboi et al., 1981, 1985). In unweaned pigs there was an age-dependent fall in lactase activity that paralleled the ontogenetic decline in villous height (Kelly et al., 1991a). Miller et al. (1986) reported that sucrase, isomaltase and lactase specific activities fell by at least 50% by five days after weaning in pigs weaned at 28 or 42 days of age. Activities of maltase II and maltase III showed no change in four-week-old pigs but increased in response to weaning at six weeks of age. Similarly, McCracken (1984), McCracken and Kelly (1984) and Kelly et al. (1990b, Kelly et al., 1991b,c) reported increases in the activities of maltase and glucoamylase when pigs were weaned onto solid diets at 14 days of age. Increases in these carbohydrases can most likely be ascribed to rapid substrate induction of these enzymes.

In several studies the decrease in villous height, the increase in crypt depth, and the loss of digestive enzyme activity after weaning, coincided with a reduced ability of weaned pigs to absorb a standard dose of D-xylose (Miller et al., 1984a,b; Hampson and Smith, 1986), alanine (Smith, 1984; Miller et al., 1986), and a solution containing glucose and electrolytes (Nabuurs et al., 1994). D-xylose is a pentose sugar that is actively absorbed across the brush-border membrane and, as with alanine, provides an assessment of the absorptive capacity of the enterocytes. However, some authors (Kelly et al., 1990b, 1991b; Pluske et al., 1996c) failed to detect a reduction in the ability of villi to absorb xylose after weaning. Miller et al. (1986) concluded that problems induced by weaning are caused more by changes in intestinal structure and specific loss of digestive enzymes rather than any gross change in absorptive function, although the data of Nabuurs et al. (1994) is in conflict with this notion.

Reasons for these discrepancies are not clear, but may be attributable to the timing of the first introduction of creep food, the quantity of diet presented and amount consumed, dietary composition, age at weaning, and differences in villous height, all of which varied between the various reports. The failure to demonstrate this in every study may not only be a reflection of the factors mentioned previously, but may also indicate that some existing tests (e.g., the xylose absorption test) are unreliable measures of absorptive function in the small intestine.

### *2.3. An alternative approach to assessing the digestive and absorptive capacity after weaning*

There is now sufficient evidence in the literature to cast doubt on the appropriateness of measures in vitro of the digestive and absorptive function in the weaned pig. Determinations of specific lactase and sucrase activity have been assumed to provide an indication of the digestive capacity in vivo (Kidder and Manners, 1980). Numerous authors have suggested the use of total gut activity as a more meaningful measure. However, the inadequacy of this index is borne out in the data of Kelly et al. (1991c) who showed that irrespective of the basis of expression, measurements of digestive enzyme activities in vitro can only provide a crude assessment of the digestive capacity and are of little value unless sup-

ported by findings in vivo of digestion and absorption.

In this respect, we (Pluske et al., 1996a) and others (Bird and Hartmann, 1994; Bird et al., 1995) at the University of Western Australia have dosed sucking and weaned pigs with physiological solutions of disaccharides and monosaccharides and traced their disappearance in the plasma over time as measures of digestion and absorption in vivo. Using this approach, Pluske et al. (1996a) found that when pigs were fed cows' fresh milk for five days after weaning there was an enhancement in the capacity of brush-border lactase and sucrase to digest physiological boluses of lactose and sucrose, respectively. Despite this increase in apparent digestive and absorptive capacity of the gut, there was a decrease in the efficiency of absorption of galactose and fructose after weaning (as measured by the 'galactose index' and the 'fructose index', which provide indices of efficiency in vivo of disaccharide digestion and monosaccharide absorption in the small intestine). This apparent anomaly may be explained by increased growth, and hence total surface area, of the small intestine after weaning such that total digestion and absorption were increased despite seeming decreases in the efficiency at which they occurred. Information such as this is impossible to obtain using current in vitro methods of assessment.

To date, this approach has only been conducted in pigs either sucking the sow or fed cows' milk after weaning. To fully assess its worth as a measure of digestive and absorptive function in vivo, experiments need to be conducted in pigs consuming solid food. Additionally, only lactose and sucrose have been used as disaccharide 'markers' of the efficacy of lactase and the sucrase–isomaltase complex. In the light of evidence presented by Kelly et al. (1991b, 1991c), it may be more beneficial to dose pigs with maltose and glucoamylose, since the enzymes responsible for their hydrolysis are likely to be more important to digestion in the weaned pig than lactase and sucrase.

### **3. Factors affecting gut structure and function after weaning**

Despite the considerable amount of research that has been conducted in this field, many questions still

remain as to the precise aetiology of these changes in gut structure and function after weaning. We have placed the factors contributing to these changes into one of five major groups, as follows:

1. enteropathogenic bacteria and their interactions in the small intestine;
2. maladaptation to the stressors at weaning;
3. the withdrawal of sows' milk at weaning;
4. the dietary change associated with weaning; and
5. cytokines as regulators of intestinal growth.

### 3.1. Enteropathogenic bacteria in the small intestine

The gastrointestinal tract of the neonate is bacteriologically sterile but, subsequently, bacteria of maternal origin and from the external environment colonise the intestine. It is well established that the indigenous microflora exerts a profound influence on both the morphological structure and on the digestive and absorptive capabilities of the gastrointestinal tract (see Kelly et al., 1992a). This is best illustrated from a comparison between conventional and germ-free, or gnotobiotic, pigs. In conventional animals the intestinal wall and lamina propria are thicker, the villi are shorter, the crypts are deeper, there is reduced activity of disaccharidases, and there is a more rapid rate of enterocyte turnover than in their gnotobiotic counterparts (see Kelly et al., 1992a). As stated by Kelly et al. (1992a), the influences exerted by both the microflora and the external environment are superimposed on the normal adaptive response of the small intestine to the dietary change that occurs at weaning.

It is common for pigs to develop a diarrhoea within 3–10 days of weaning. The condition is associated with proliferation of  $\beta$ -haemolytic *Escherichia coli* (*E. coli*) in the proximal small intestine of affected pigs, and has been well documented throughout the world as a cause of significant economic loss on affected piggeries. Although specific serotypes of *E. coli* have a central role in the aetiology of post-weaning diarrhoea (PWD), the condition is complex and multi-faceted. Diarrhoea may also be caused by rotavirus (Lecce et al., 1983), and the role of this microbe in the aetiology of PWD should also be considered.

Predisposition to infection by enterotoxigenic microbes involves a number of factors, and these have

been described by McCracken and Kelly (1993). It is impossible to define the extent to which these factors interact as they happen simultaneously around the time of weaning. A strong association exists between colonisation of the small intestine of newly-weaned pigs by haemolytic *E. coli* and the occurrence of diarrhoea, however oral dosing with these same bacteria often fails to induce disease (see Hampson, 1994). Similarly, Hampson et al. (1985) and Nabuurs et al. (1993a) reported that rotavirus and enterotoxigenic *E. coli* (ETEC) were generally detected when pigs had diarrhoea, but were also encountered in normal faeces from healthy pigs. Additionally, haemolytic strains of *E. coli* proliferate in the small intestine of healthy pigs following weaning, although in lower numbers than their counterparts which develop diarrhoea (Kenworthy and Crabb, 1963; Svendsen et al., 1977).

Nabuurs et al. (1993b) reported that pigs taken from Dutch herds having a long history of PWD had shorter villi and deeper crypts than their counterparts from a specific-pathogen-free herd. In addition, pigs dying after weaning from herds having diarrhoea had shorter villi and deeper crypts than in pigs of those herds without deaths. From these data, Nabuurs et al. (1993b) suggested that villous height and crypt depth may influence the pathogenesis of diarrhoea after weaning. They postulated that the relationship between intestinal architecture and diarrhoea may stem from the function of villous enterocytes and crypt cells, since shorter villi and deeper crypts have fewer absorptive and more secretory cells that causes decreased absorption but increased secretion. A reduction in digestion and absorption would encourage the development of an osmotic diarrhoea, whilst unabsorbed dietary material may act as a substrate for ETEC in the gut (Hampson, 1994). Changes in villous enterocyte populations may also expose new receptors such as for the strains of ETEC that do not possess K88 fimbria (Nagy et al., 1992).

To support this argument, Nabuurs et al. (1994) reported that the net absorption of fluid, sodium, potassium and chloride in weaned and unweaned pigs was less in segments infected with ETEC than in those not infected (Fig. 2). In uninfected weaned pigs, absorption was less on days 4, 7 and 14 after weaning than in similarly-treated gut segments obtained from unweaned pigs. These authors reported a

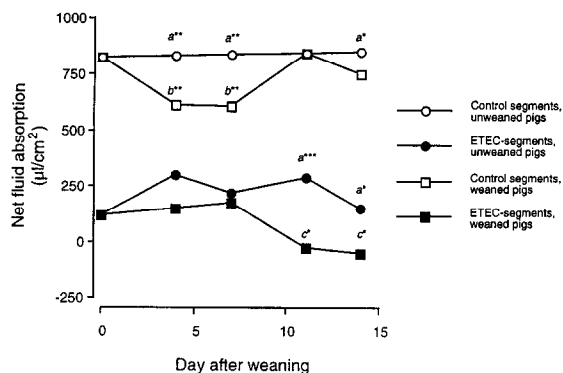


Fig. 2. Mean net absorption of fluid in uninfected and ETEC-infected segments of small intestine in unweaned and weaned pigs on the day of weaning, and on days 4, 7, 11 and 14 thereafter (a: net absorption greater than in weaned pigs on same day; b: net absorption less than on day 11 in similarly-treated pigs; c: net absorption less than on days 4 and 7 in similarly-treated pigs. \*  $P < 0.05$ ; \*\*  $P < 0.01$ ; \*\*\*  $P < 0.001$ ) (redrawn from Nabuurs et al., 1994).

negative correlation ( $r = -0.95$ ,  $P < 0.02$ ) in ETEC-infected weaned pigs between net absorption and villous height, an observation that may be explained by the presence of immature enterocytes on the villi (Symons, 1965; Wild and Murray, 1989) and the enhanced susceptibility of immature enterocytes to bacterial toxins (Cohen et al., 1986; Chu and Walker, 1993). Moreover, Nabuurs et al. (1994) found that the net absorption of the entire small intestine was reduced in ETEC-infected weaned pigs compared to ETEC-infected unweaned pigs.

In an earlier study, where the combined effects of weaning and rotavirus infection in gnotobiotic pigs were examined in relation to gut structure and function, Hall et al. (1989) reported no evidence that inoculation with rotavirus at the time of introduction of a solid diet caused persistent damage, both in terms of gut architecture and brush-border enzyme activities, to the small intestine. Any damage, which occurred in the mid and distal portions of the small intestine, was patchy and short lived, and had no detrimental effects on growth rate. Hall et al. (1989) concluded that rotavirus alone was not responsible for changes in gut structure and function after weaning, but cautioned that the possibility of interactions between rotavirus and other microorganisms (e.g., *E. coli*) present in conventionally-reared weaned pigs should not be discounted.

Again using gnotobiotic pigs, Hall and Byrne (1989) showed a direct effect of dietary change, uncomplicated with pathogens, on the structure and function of the small intestine after weaning. The mucosa of the small intestine was damaged when gnotobiotic pigs were weaned onto a pelleted meal diet, with villous height, crypt depth, crypt-cell production rate, and activities of lactase and sucrase being reduced. Hall and Byrne (1989) suggested that the cause of villous atrophy was a slowing of new cells produced in the crypts and not an accelerated loss of mature enterocytes from the villi surface. Damage to the small intestine was associated with reduced weight gain over a three-week period, and diarrhoea was not observed. These authors suggested that reduced food consumption by pigs after weaning, leading to sub-optimal protein and energy intakes, may reduce crypt-cell production rate. This finding concurs with those of Kelly et al. (1991c) and Pluske et al. (1996c) who reported reduced crypt depth in pigs fed a reduced amount of food after weaning. An anomaly exists, however, because in many situations crypt depth increases after weaning even when food intake per se is insufficient to meet the daily maintenance requirement of the pig. For example, Miller et al. (1986) reported a reduced crypt depth in pigs that were weaned into a 'clean' environment compared to pigs weaned into a 'dirty' environment when fed at levels below maintenance. It seems likely, therefore, that a degree of interaction occurs between the microflora, the dietary change at weaning, and other factors such as the environment to determine the rate of crypt-cell production.

In conclusion, the precise role that ETEC plays in causing and (or) predisposing changes in villous height and crypt depth in the small intestine after weaning is difficult to assess since these alterations are the result of a number of changes occurring at weaning. It is not possible to say at times, therefore, whether proliferation of *E. coli* in the gut of the weaned pig is a cause or an effect of the general malaise associated with the weaning process per se.

### 3.1.1. Interactions between bacteria, the intestinal mucosa, and diet

Components of the diet and the gut microflora are in intimate contact within the intestinal tract, and it is likely that dietary composition may influence the

carbohydrate structures of the mucosal and mucin glycoconjugates with marked consequences for the adhering microflora and to the gut itself (Kelly et al., 1992a). In the suckling pig, exposure to components of colostrum and milk (e.g. secretory immunoglobulins such as sIgA, lactoferrin, lysozyme, lymphocytes, phagocytes, oligosaccharides) is likely to alter bacteria growth but, when these compounds disappear at weaning, not only will this make the pig more vulnerable to infection by opportunistic and other pathogens, but it will most likely alter the gut morphology and function. Kelly et al. (1992a, 1994) and Kelly (1994) provide descriptions of the subtle changes and interactions that occur between diet and bacterial receptors in the gut, and this will not be reiterated here.

### 3.2. *Maladaptation to the stressors at weaning*

There is little information relating effects of psychological stress (e.g. mixing and moving) imposed on pigs at weaning to changes in gut structure and function, and studies that have been reported in the literature refer only to pigs suffering from 'wasting pig syndrome'. This is a term that describes 'wasting' or unthrifty pigs that do not recover from the initial growth check associated with weaning.

Hall et al. (1983) reported gut atrophy and reductions in brush-border enzyme activity in unthrifty pigs after weaning, but commented that five weeks later, gut structure and function were similar to normal pigs of the same age. Albinsson and Andersson (1990) reported that in pigs weaned at five weeks of age but deemed to be suffering 'wasting pig syndrome' at 10 weeks of age, there was a reduced rate of crypt-cell proliferation, a reduced villous height, and a decreased alkaline phosphatase activity in ileal mucosa compared to healthy counterparts. Furthermore, administration of amperozide, a psychotropic drug having specific cerebral effects on aggressive behaviour, to wasting pigs was shown to normalise villous height and plasma alkaline phosphatase activity, of which the latter has been used as a general indicator of growth rate. Similarly, Kyriakis and Andersson (1989) reported a therapeutic effect of amperozide on wasting pigs that enhanced subsequent growth rate, and Björk (1989) found that amperozide led to improved performance after wean-

ing, an effect attributed to amelioration of 'gastro-intestinal dysfunction' associated with weaning.

Despite these findings, Albinsson and Andersson (1990) failed to establish a relationship between the stress response, plasma levels of cortisol and cortisol binding globulin, and growth rate in wasting pigs. Similarly, a decreased villous height in wasting pigs was not associated with increased levels of either 11-deoxycortisol or cortisol. These authors failed to measure the endocrine indices of adrenal activity following amperozide injection, so it is difficult to surmise whether the reported improvement in villous height was a direct consequence of amperozide per se or was caused by other factors associated with amperozide administration, such as a concomitant increase in food intake. This concurs with the authors' proposition that disturbances in feeding patterns and the aggressive interactions caused by mixing unfamiliar pigs at weaning, cause wasting pig syndrome.

In this regard, Pluske and Williams (1996) commented that changes in gut structure and function purported to be a consequence of psychological stress imposed at weaning (i.e. separation from the sow, moving and mixing) are most likely confounded with the low levels of voluntary food intake seen at this time. Partial confirmation of this comes from the recent work of Beers-Schreurs et al. (1995), who reported that pigs separated from the sow at weaning and fed sows' milk had villi and crypts of similar height and depth, respectively, to pigs left sucking the sow. So, although psychological stress most likely causes 'wasting pig syndrome', we suggest that factors responsible for the changes in gut structure and function are more complex and involve additional factors, such as the amount of food pigs eat after weaning, rather than any direct effects of psychological stress per se. This notion is supported further by the data of McCracken et al. (1995) and Spurlock et al. (1996), who reported diet-independent metabolic changes in interleukin 1 and acute-phase proteins resulting from the stressors imposed at weaning.

### 3.3. *The withdrawal of sows' milk at weaning*

There is increasing interest in the important role that colostrum- and milk-borne growth factors, hormones and other bioactive substances may play in

postnatal differentiation and development of the small intestine of the pig. This is especially pertinent to the weaned pig because the source of these compounds, i.e. sows' milk, is removed abruptly at weaning, leaving the small intestinal epithelium devoid of these compounds. This is likely to have marked effects on the processes regulating cell growth, cell differentiation and cell function in the small intestine. Bioactive compounds implicated in small intestinal development in the young pig include epidermal growth factor (EGF), polyamines, insulin, and the insulin-like growth factors.

### 3.3.1. Epidermal growth factor (EGF)

One of the best characterised growth factor receptors in the small intestine of the pig is that for EGF (Jaeger and Lamar, 1992; Kelly et al., 1992b). Binding of exogenous EGF to receptors increases during suckling to reach detectable levels after weaning (Kelly et al., 1992b) and, as stated by Kelly et al. (1992a), it is possible that the ontogenic increase in receptor levels of EGF during suckling is related to the presence of endogenous milk factors which either occupy, modify, or downregulate the EGF receptor. The epidermal growth factor is present in high concentrations in colostrum and sows' milk (Odle et al., 1996), so the continued presence of intestinal receptors for EGF in young pigs provides support for the role of EGF as a modulator of gut growth and development. A review of the intestinal effects of EGF in the young pig is provided by Odle et al. (1996).

### 3.3.2. Polyamines

Polyamines (putrescine, spermidine and spermine) and the key enzymes controlling their synthesis (ornithine decarboxylase and S-adenosylmethionine decarboxylase) are critical for postnatal cellular proliferation and differentiation (Kelly, 1994). Polyamines are found in high concentrations in porcine milk and intestinal tissue (Kelly et al., 1991d), and their total absence from the diet at weaning may also be responsible for some of the observed changes in gut structure and function seen after weaning. This is especially relevant since it has been shown that hormones, growth factors and other nutrients that stimulate intestinal differentiation also increase the intracellular concentration of polyamine

(Kelly, 1994). For example, Olanrewaja et al. (1992) demonstrated that the trophic action of IGF-1 was dependent on polyamine biosynthesis and uptake. Interestingly, Grant et al. (1990b) showed an ameliorating effect of feeding polyamines to weaner pigs on gut structure and function, effects also observed in calves (Grant et al., 1990a) and week-old chicks (Mogridge et al., 1996) that were fed soyabeans.

### 3.3.3. Insulin and insulin-like growth factors

Insulin and insulin-like growth factor (IGF) receptors have been described in the intestinal epithelium of various species, including the pig. A review of the concentrations of these two compounds in serum and sow mammary secretions, together with their gastrointestinal effects when administered exogenously to young pigs, can be found in Odle et al. (1996).

Of recent interest is the role of the structurally-homologous growth factors, IGF-1 and IGF-II, in gut growth and development of the young pig. This is not only because sucking pigs ingest physiologically significant amounts of IGF-1 via colostrum and milk (Simmen et al., 1988, 1990), but also relates to the finding that exogenous IGF-1 (or analogue) treatment does not feedback in a negative manner to inhibit endogenous IGF-1 secretion, thereby allowing the growth-promoting properties of this hormone to be expressed (Schoknecht et al., 1993).

Unpublished work from Walton and Dunshea (see Dunshea and Walton, 1995) shows the interactions between nutrient intake (manipulated through establishing litter sizes of six and 12 pigs on the sow) and infusion of an IGF-1 analogue, LR<sup>3</sup>IGF-1, in sucking pigs. Subcutaneous infusion of LR<sup>3</sup>IGF-1 not only increased piglet growth rate in late lactation, but also increased the growth of the small intestine, spleen and pancreas (Table 1). Similar results have been reported by Burrin et al. (1995), in which pigs deprived of colostrum but which received milk replacer fortified with IGF-1 for four days had greater intestinal weight (28%) and higher villi in the jejunum (78%) than their control counterparts. Xu et al. (1994) found no effect on piglet growth rate but an increase in pancreas size after pigs received infant formula supplemented with IGF-1 for 24 hours, but this length of time may have been too short to see a growth response and changes in small intestinal growth.



Table 1

Effect of LR<sup>3</sup>IGF-1 subcutaneous infusion on growth performance and organ size at 27 days of age in sucking pigs from litters of six or 12 pigs (after Walton and Dunshea, unpublished data)

|                                   | Litter size |                       |         |                       |                  | P-value        |                |
|-----------------------------------|-------------|-----------------------|---------|-----------------------|------------------|----------------|----------------|
|                                   | Six         |                       | Twelve  |                       |                  | L <sup>b</sup> | T <sup>b</sup> |
|                                   | Control     | LR <sup>3</sup> IGF-1 | Control | LR <sup>3</sup> IGF-1 | SED <sup>a</sup> |                |                |
| Average daily gain, g/day         |             |                       |         |                       |                  |                |                |
| Day 0–27                          | 299         | 304                   | 187     | 199                   | 19.0             | ***            | n.s.           |
| Day 18–27                         | 294         | 325                   | 114     | 167                   | 32.0             | ***            | *              |
| Organ weight at 27 days of age, g |             |                       |         |                       |                  |                |                |
| Small intestine                   | 359         | 373                   | 247     | 311                   | 34.0             | **             | *              |
| Liver                             | 263         | 312                   | 168     | 221                   | 23.0             | ***            | **             |
| Spleen                            | 29          | 53                    | 16      | 40                    | 6.3              | *              | ***            |

<sup>a</sup>SED: standard error of difference between interaction means.

<sup>b</sup>L, litter size; T, LR<sup>3</sup>IGF-1 treatment.

n.s.: not significant; \*  $P < 0.05$ ; \*\*  $P < 0.01$ ; \*\*\*  $P < 0.001$ .

In the light of these data, and given that the content of growth factors in sows' milk declines with advancing lactation (Donovan et al., 1994), an opportunity may exist to enhance gut growth and development through supplementation of the newly-weaned pig with exogenous growth factors. As argued by Dunshea and Walton (1995), this is particularly pertinent in the case of the weaner pig since (a) its gut is relatively 'immature' at weaning, (b) the pig suffers a growth check, and (c) the gut of the newly-weaned pig is often challenged by enteropathogenic bacteria. In this instance, the increased spleen size seen in the work of Walton and Dunshea (Table 1) may reflect an enhanced immune response, although it may also be a result of tissue-specific tropism of LR<sup>3</sup>IGF-1. The results of the IGF-1-analogue infusion studies lend strong support for a role of IGF-1 and (or) its analogues to stimulate growth and visceral development in the weaned pig, and is an area worthy of future investigation as a means of overcoming the post-weaning growth check and enhancing the digestive and absorptive capacity of the newly-weaned pig.

#### 3.3.4. A role for L-glutamine in changes to gut structure and function after weaning?

Changes in gut structure and function after weaning may also be influenced by availability of the amino acid L-glutamine. Glutamine is the principal respiratory fuel for gut enterocytes and provides amide nitrogen to support nucleotide biosynthesis

(Windmueller, 1982). Numerous studies using mainly dogs and rodents have demonstrated that glutamine is required by villous enterocytes to support enterocyte metabolism and the structure and function of the small intestine (see Souba, 1991, 1993). Provision of oral glutamine stimulates the net uptake of glutamine by enhancing the brush-border transport rates (Salloum et al., 1990), and supports mucosal growth by stimulating the activity of glutaminase (Klimberg et al., 1990; Salloum et al., 1989). At a time when the piglet's supply of maternal glutamine disappears, and the endogenous supply of glutamine from muscle and plasma to the gut epithelium may be inadequate to maintain villous integrity, supplementation of weaner diets with synthetic glutamine offers a means of enhancing the structure and function of the gut after weaning.

In sows' milk, Wu and Knabe (1994) reported that glutamine was the most abundant free and protein-bound amino acid at days 22 and 29 of lactation. Wu and Knabe (1993) also measured glutamine metabolism in isolated enterocytes and found two- and ten-fold increases in the rate of oxidation of glutamine to CO<sub>2</sub> in enterocytes from 29-day-old weaned pigs compared to 21-day-old sucking pigs. These data suggest that glutamine may serve as an increasingly important energy substrate for the enterocytes of weaned pigs, and it is tempting to hypothesise that glutamine is a conditionally essential amino acid for the weaned pig. Several lines of evidence support this notion.

Wu et al. (1996) reported that the addition of 1% glutamine to a corn-soyabean diet prevented villous atrophy in the jejunum on the seventh day after weaning. Ayonrinde et al. (1995a,b) fortified a conventional cereal-based weaner diet with either 4% glutamine or 4% glycine and, on the fifth day after weaning, slaughtered all pigs. Pigs fed the diet containing exogenous glutamine had greater plasma glutamine concentrations, and higher villi and deeper crypts consistent with amelioration of villous atrophy and stimulation of crypt-cell production rate (Fig. 3). Furthermore, other indices of jejunal and ileal integrity such as DNA concentration and mucosal protein content were enhanced by feeding glutamine (Fig. 3). McBurney et al. (1994) found that consumption of  $\approx 6.3$  g free glutamine/day, which was added to a conventional weaner diet, maintained plasma and muscle free glutamine concentrations similar to those measured in 21-day-old sucking pigs. Pluske et al. (1996b) reported a linear increase in crypt depth with increasing glutamine intake in pigs fed ewes' whole milk for five days after weaning, indicative of an increase in glutamine metabolism in the epithelium. An increase in glutamine transport across the enterocytes increases the activity of mitochondrial glutaminase (Klimberg et al., 1990; Salloum et al., 1990; Ayonrinde et al., 1995b). This would accelerate the hydrolysis of glutamine into products that can directly enter the TCA cycle and

generate ATP. This energy could be used to support cell division in the proliferative zone in the crypts of Lieberkühn (Newsholme et al., 1985; Klimberg et al., 1990), since glutamine is a requisite substrate for DNA biosynthesis *de novo*.

Finally, there is evidence in the pig that luminal glutamine is beneficial for the maintenance of normal mucosal permeability in states of enteric infection (Dugan and McBurney, 1995). As discussed previously, the health of the newly-weaned pig is often compromised by proliferation of enteropathogenic bacteria that cause marked structural and functional changes to the small intestine. In this context, the role of glutamine in the newly-weaned pig, especially under conditions of enteric infection where gut epithelial and gut-associated lymphoid tissue requirements for glutamine would be expected to increase (Wu et al., 1991), may be worthy of future investigation.

### 3.4. Dietary changes at weaning

#### 3.4.1. Type of diet fed after weaning

Deprez et al. (1987) compared a pelleted diet with the same diet fed in slurry form, and recorded higher villi on days eight and 11 after weaning when pigs were fed the slurry. Villous height may have been maintained after weaning because the digesta from a pelleted diet may be more abrasive than that from liquid diets, which could decrease the villous height by increasing the shedding of enterocytes. Alternatively, the higher villi recorded by these workers in pigs fed the slurry diet may be a reflection of their higher level of energy intake.

Two lines of evidence support this notion. First, Partridge et al. (1992) showed that weaned pigs fed a dry, solid diet in slurry form consumed 13% more food ( $P < 0.05$ ) and grew 11% faster ( $P < 0.01$ ) than pigs fed the same diet in pelleted form. Second, we (Pluske et al., 1996b,c) and others (Kelly et al., 1991c; Beers-Schreurs et al., 1995) have demonstrated that villi are higher in pigs that consume more food. A better comparison of gut structure and function in pigs fed the same diet in solid or slurry form may occur, therefore, if energy intake was equalized between the two groups.

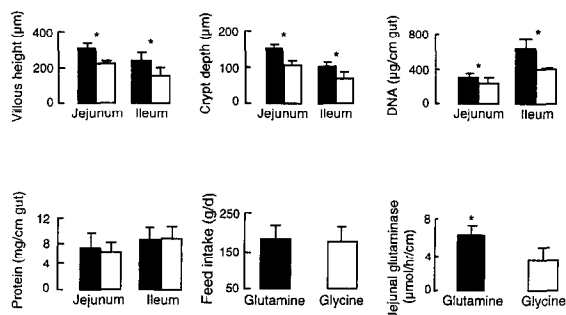


Fig. 3. Villous height and crypt depth in the jejunum, ileum and DNA concentration, mucosal protein concentration, voluntary food intake, and jejunal glutaminase activity of pigs fed solid diets supplemented with glutamine (■) or glycine (□) for five days after weaning (mean  $\pm$  SEM). \* Differences between treatments were significant (redrawn from Ayonrinde et al., 1995b).

Another study examining the influence of weaning diet and energy intake on gut structure was conducted by Beers-Schreurs et al. (1995). These workers weaned pigs at 28 days of age and offered them one of three diets: (i) sows' milk on an ad libitum basis, (ii) a commercial diet on an ad libitum basis, and (iii) sows' milk corresponding to the same amount of energy that the pigs eating the commercial diet consumed the day before. Pigs in groups (ii) and (iii) consumed less food and had smaller villi than pigs fed sows' milk on an ad libitum basis, suggesting that reduced energy intake – independent of diet type – is a major cause of villous atrophy after weaning.

#### 3.4.2. *Transient hypersensitivity to food components in the diet*

It has been suggested that the morphological changes observed after weaning are the product of a transient (delayed) hypersensitivity to antigenic components of the diet (see reviews by Miller et al., 1991; Lallès et al., 1993; Lallès and Salmon, 1994). This has become one of the most debated issues in the development of feeding strategies for the weaned pig, since the proposition has been made that the immunopathological damage to the small intestine would vary in severity between abruptly weaned, sensitized (i.e. low creep food intake before weaning) pigs, and immunologically tolerant (i.e. adequate creep food intake before weaning) pigs (see Hampson, 1987). According to this hypothesis, a high intake of creep food before weaning favours immune tolerance whereas a low intake primes immune tolerance and predisposes the weaned piglet to diarrhoea. However, data to support this hypothesis, in both feeding experiments and studies where infection has been induced using enterotoxigenic strains of *E. coli*, is equivocal (see Hampson (1987), Kelly et al. (1992a) and Lallès and Salmon (1994) for further discussions), with some authors even reporting an increased incidence of diarrhoea associated with a high consumption of dry food prior to weaning (Barnett et al., 1989).

#### 3.4.3. *Soya proteins in diets for weaned pigs*

There is a body of evidence suggesting that soyabean meal included in diets for weaner pigs is antigenic and stimulates a localised immune response, although this is not always related to the

incidence of diarrhoea after weaning (Li et al., 1990; Kelly et al., 1990a). This response is thought to be caused by immunologically-active soyabean proteins such as glycinin and  $\beta$ -conglycinin that cause a delayed hypersensitivity reaction. For example, Li et al. (1990, 1991a,b) reported a decreased villous height, an increased crypt depth, increased serum anti-soy IgG titres, an increased skin-fold thickness after intradermal injection of soy protein, and a decreased xylose absorption, in pigs sensitized (i.e. orally infused with soyabean meal) during suckling and then fed soyabean meal after weaning in comparison to pigs sensitized with dried skim-milk powder. Li et al. (1991a,b) and Dréau et al. (1994) also showed differences in these parameters between different soyabean products.

In contrast, Kelly et al. (1990b) reported no differences in villous height, crypt depth and intraepithelial lymphocyte count in weaned pigs given either no creep food, a low level of creep food, or a high level of creep food. The creep food contained 6.5% soyabean meal (48% crude protein), so the total amount of soyabean meal given in the high group was similar to that given to pigs in the study reported by Li et al. (1990). A similar finding was reported by Hampson et al. (1988).

Despite these findings, antigenic soya protein appears to exert stronger effects on gut structure in the first two weeks after weaning than skim milk powder or low antigenic material in both normal (Dunsford et al., 1989; Li et al., 1990; Miller et al., 1991; Dréau et al., 1994) and gnotobiotic (Hall and Byrne, 1989; Ratcliffe et al., 1989) pigs. Growth setbacks following inclusion of soyabean meal in diets have also been reported (Li et al., 1990, 1991a,b), although any long-lasting effects of soyabean meal on subsequent growth rate are not apparent, presumably because pigs become systemically tolerant to soy proteins a few weeks after weaning (Stokes et al., 1987; Barnett et al., 1989; Heppell et al., 1989; Wilson et al., 1989). Therefore, the exclusion of soyabean meal from starter diets to overcome any deleterious effects on gut structure and function may increase growth initially, but its inclusion at a later stage (e.g. 14 days after weaning) causes a growth penalty similar to that observed when soyabean meal is included in diets from the time of weaning (Friesen et al., 1992).

A salient aspect of these experiments, except that of Kelly et al. (1990b), was the lack of consideration given for the possible interaction between soyabean protein and the level of feeding per se on gut structure after weaning. In the studies of Li et al. (1990, 1991a,b) and Dréau et al. (1994), small doses of soyabean protein were gavage-fed during suckling so that intake was controlled precisely, but pigs were allowed free access to food after weaning. Pigs in these studies grew poorly. For example, Dréau et al. (1994) reported that in the first week following weaning at 21 days of age, pig growth rate was between  $-21$  g/day and  $100$  g/day. No estimates of voluntary food intake were provided by these authors, but undoubtedly these pigs were not eating enough food to cover their maintenance requirement for energy. This scenario is synonymous to short-term starvation which causes villous atrophy (Pluske et al., 1996c). What may be confounded in these studies, therefore, is the precise contribution of antigenic soyabean protein on villous height and crypt depth since these indices of gut function are also influenced by level of feeding (Kelly et al., 1991c; Pluske et al., 1996b,c).

Nevertheless, densities of T and B lymphocytes are increased in small intestinal tissue in pigs exposed to highly-antigenic, heated, soyabean flour, an effect which could have directly influenced enterocyte kinetics (Dréau et al., 1995). In addition, feeding soyabean meal may have quite important effects on electrolyte secretion in the small intestine (Naburs, 1986) and, in young calves, has been shown to have immediate toxic effects on small intestinal morphology (Kilshaw and Slade, 1982).

#### 3.4.4. *Anti-nutritional factors in diets*

In addition to the anti-nutritional effects associated with feeding soyabean meal, it is well recognised that other factors such as lectins, tannins, and  $\alpha$ -amylase inhibitors can also decrease production due to their effects on gut structure and function. Recent reviews on the presence, distribution and physiological effects of anti-nutritional factors in the gut of pigs have been presented by Huisman and Jansman (1991), Huisman and Tolman (1992) and Lallès et al. (1993), and will not be covered in this review.

#### 3.4.5. *Role of enteroglucagon and short-chain fatty acids*

A consequence of the dietary change at weaning is an increase in the amount of material which enters the large intestine, with a concomitant increase in the rate of microbial fermentation. An increase in nutrient load reaching the lower part of the small intestine stimulates the release of the polypeptide hormone enteroglucagon from endocrine cells located in the mucosa (Al-Mukhtar et al., 1982; Kelly et al., 1991c). Several authors have correlated crypt-cell production rate with levels of enteroglucagon in the plasma (Al-Mukhtar et al., 1982; Goodlad et al., 1983). Since receptors for enteroglucagon show a proximal to distal increase along the length of the small intestine (Bloom, 1980), and its release into the circulation is stimulated by the presence of digesta in the ileum, the presence of enteroglucagon cells in the distal part of the small intestine would appear to be well positioned to monitor digesta for the presence of unabsorbed nutrients. This could then operate a feedback control of epithelial cell proliferation (Sagor et al., 1982).

To our knowledge, only one study has measured plasma enteroglucagon levels in the weaner pig. Kelly et al. (1991c) found that pigs fed continuously as opposed to restrictedly for five days after weaning had higher ( $P = 0.055$ ) levels of plasma (N-C)-terminal enteroglucagon, which is a direct measure of gut hormone concentration. Increases in the weight of the small intestine, the mucosa of the small intestine, and villous height and crypt depth reported by Kelly et al. (1991c) in pigs fed continuously suggests a trophic role for enteroglucagon. Enhanced secretion of this hormone may be the mechanism that alters intestinal adaptation in response to the level of nutrient intake (Kelly et al., 1991c).

Alternatively, deeper crypts may be attributable in part to the increased production of short-chain volatile fatty acids in the large intestine that, in turn, stimulate crypt-cell production rate in the small intestine (Sakata, 1987; Yajima and Sakata, 1987; Goodlad et al., 1989). Support for this notion in the pig literature comes from a report by Jin et al. (1994). These authors found that growing pigs fed a high-fibre diet had a greater rate of crypt-cell proliferation in the jejunum and colon than pigs fed the low-fibre diet. This equates to an increased rate of turnover of

intestinal mucosal cells, and suggests that dietary fibre reaching the hind-gut may also be affecting gut structure in the small intestine via the influence of volatile fatty acids.

#### 3.4.6. Voluntary food intake after weaning and its effects on gut structure and function

Voluntary food intake in the post-weaning period is both low and extremely variable, with pigs often failing to consume sufficient food to cover their energy requirement for maintenance (Bark et al., 1986). Le Dividich and Herpin (1994) summarized several data sets and concluded that the metabolisable energy (ME) requirement for maintenance is not met until the fifth day after weaning, with pre-weaning ME intake not being attained until the end of the second week following weaning (Fig. 4).

As one of the most potent stimuli of intestinal proliferation is the presence of food in the intestinal lumen or, more specifically, nutrient flow along the small intestine (Diamond and Karasov, 1983), the absence of nutrients from the gut lumen such as that which occurs after weaning will have marked effects on the rate of cell differentiation and cell turnover. A large body of evidence has now accumulated indicating that the oral intake of food and its physical presence in the gastrointestinal tract per se are necessary for structural and functional maintenance of the intestinal mucosa (see Kelly et al., 1992a). The presence of food in the gastrointestinal tract has

direct and indirect effects on epithelial cell proliferation (Johnson, 1987). It is well recognised, for example, that the exclusion of nutrients from the lumen of the small intestine either by starvation (e.g. McNeill and Hamilton, 1971; Altmann, 1972), dietary restriction (e.g. Núñez et al., 1995), or intravenous feeding (e.g. Goodlad et al., 1992), results in villous atrophy and a decrease in crypt-cell production rate. Since these changes have been reported in the gut of the newly-weaned pig, it is likely that luminal nutrition plays a strong role in the integrity of the structure and function of the small intestine after weaning.

The major effect of starvation and re-feeding is to increase and decrease, respectively, the duration of the cell-cycle time, or  $T_c$ , in the crypts of Lieberkühn (Al-Dewachi et al., 1975). An increase in cell-cycle time is caused by extending the  $G_1$  and S phases of the mitotic cycle (Koga and Kimura, 1980). Numerous workers have also reported an increase in cell-cycle time, or a decrease in the production of crypt cells, associated with starvation (e.g. Altmann, 1972; Al-Mukhtar et al., 1982; Goodlad et al., 1983; Goodlad et al., 1988). Re-feeding for as little as 9–12 hours causes a reduction in cell-cycle time and a general increase in cell proliferation in the crypts (Al-Dewachi et al., 1975). Goodlad and Wright (1984) starved adult rats for 24 hours and found a reduction in crypt-cell production rate along the entire length of the gut. After re-feeding for nine hours there was a marked increase in the production rate of crypt cells, especially in the proximal small intestine. Goodlad and Wright (1984) concluded that after re-feeding, cell migration from crypt to villus is not immediately dependent upon cell proliferation, but may be a response to the presence of nutrients in the lumen stimulating cell migration directly to produce an immediate increase in digestive and absorptive capacity.

The reduction in live weight gain caused by decreased food intake after weaning reduces fasting heat production (Koong et al., 1982). Since heat production is associated with protein synthesis (Webster, 1980, 1981), and this is most active in the digestive tract (Pekas and Wray, 1991), any further reductions in food intake would be expected to reduce the rate of cell production and decrease cell renewal in the small intestine. Furthermore, Koong and Ferrell (1990) and Pekas and Wray (1991)

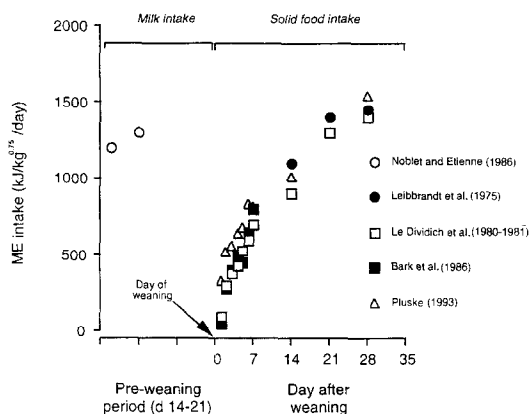


Fig. 4. The voluntary food intake of pigs before and after weaning (expressed as metabolisable energy (ME) intake per metabolic kilogram) (adapted from Le Dividich, 1981; Le Dividich et al., 1980; Leibbrandt et al., 1975; Noblet and Etienne, 1986; Le Dividich and Herpin, 1994).

demonstrated that the energy expenditure of the small intestine varies directly with fasting heat production in the growing pig under different nutritional regimens. If the gut mucosa responds directly to the level of energy intake, it is likely, therefore, that the structure and function of the small intestine also depends on the level of intake.

In many experiments, it has not been possible to establish the likely contribution of food intake per se on morphological and biochemical alterations in the gut after weaning because researchers have usually failed to quantify the amount of food consumed. As a consequence, they have been unable to relate food intake to the histological and enzymological changes observed. Where estimates of food intake have been documented, the reported changes in gut structure and function are most likely confounded with the period of low food intake that occurs immediately after weaning.

The first workers to recognise that low food intake in the period immediately after weaning may be responsible for changes in gut structure and function in the pig were Kelly et al. (1984) and McCracken and Kelly (1984). These authors adopted the technique of gastric intubation as a means of regulating food consumption and reducing the variation in food intake after weaning. Commensurate with a similar report by McCracken (1984), these workers suggested that mucosal atrophy after weaning may be related more to the lack of a continuous supply of substrate than to any antigenicity in the diet or to inherently low levels of digestive enzyme activity.

Despite these earlier reports, only Kelly et al. (1991c) and Pluske et al. (1996c) have studied the effects of different levels of food intake on the digestive and absorptive development of the weaned piglet. Kelly et al. (1991c) fed by stomach tube a low or high amount of a cereal-based diet to pigs weaned at 14 days of age for the first five days after weaning. Pigs fed less food showed villous atrophy and decreased crypt depth at all sites along the small intestine compared to pigs fed a higher quantity of food (Fig. 5).

Estimates of brush-border enzyme activity and xylose absorption failed to corroborate these marked changes in gut structure induced by differential feeding (Table 2), a result also supported by the findings of Pluske et al. (1996c). Kelly et al. (1991c) reported

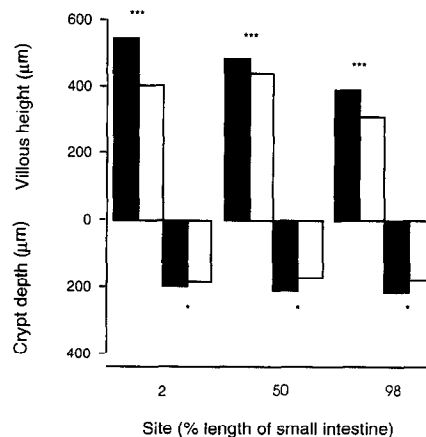


Fig. 5. Villous height and crypt depth at 2, 50 and 98% of the pyloro-ileal intestinal length in pigs weaned at 14 days of age and given either a high (■) or low (□) nutrient supply. \*\*\*  $P < 0.001$  for villous height, high *v.* low feeding (SEM = 15.2); \*  $P = 0.016$  for crypt depth, high *v.* low feeding (SEM = 7.4). Pigs fed a *high* amount of food were each given 150, 175, 200, 225 and 250 g, and pigs fed a *low* amount of food were each given 0, 25, 50, 75 and 100 g food, each day for five days after weaning (redrawn from Kelly et al., 1991c).

that feeding more food increased the total activities of maltase and glucoamylase in accordance with substrate-induction of these carbohydrases.

Pluske (1993) reasoned that if the nutritional stress of interrupted intake at weaning could be overcome, then the transition from sows' milk to solid food would be less traumatic and piglet growth would increase. Since an increase in food intake in the immediate post-weaning period is likely to exert potent stimulatory effects on mucosal growth and function, this may preserve the integrity of the small intestine and promote growth through an enhancement and (or) preservation of digestive and absorptive capacity.

By coaxing pigs to drink cows' fresh milk immediately after weaning at two-hourly intervals, Pluske et al. (1996b) demonstrated that villous height and crypt depth could be maintained in pigs after weaning by feeding a milk liquid diet. This suggests that the balance between cell loss from the villi and cell production in the crypts was preserved under these feeding conditions. Pluske et al. (1996c) also demonstrated that when pigs were weaned onto cows' fresh milk and offered three levels of energy intake (i.e.

Table 2

Mean values measured over five sites along the small intestine for lactase (EC 3.2.1.23), sucrase (EC 3.2.1.48), maltase (EC 3.2.1.20) and glucoamylase (EC 3.2.1.3) activities, and serum xylose concentration, of pigs weaned at 14 days of age and given either continuous or restricted nutrient supply for five days (after Kelly et al., 1991c)

|                                               | Feeding level <sup>a</sup> |            | SEM <sup>b</sup> | P-value |
|-----------------------------------------------|----------------------------|------------|------------------|---------|
|                                               | Continuous                 | Restricted |                  |         |
| Lactase                                       |                            |            |                  |         |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ protein | 66                         | 86         | 9.9              | 0.15    |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ mucosa  | 10                         | 11         | 1.3              | 0.80    |
| mol/day                                       | 0.8                        | 0.7        | 0.09             | 0.60    |
| Sucrase                                       |                            |            |                  |         |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ protein | 74                         | 100        | 9.7              | 0.08    |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ mucosa  | 12                         | 13         | 1.2              | 0.65    |
| mol/day                                       | 0.9                        | 0.9        | 0.10             | 0.75    |
| Maltase                                       |                            |            |                  |         |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ protein | 24                         | 23         | 2.2              | 0.71    |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ mucosa  | 4                          | 3          | 0.4              | 0.04    |
| mol/day                                       | 0.3                        | 0.2        | 0.03             | 0.01    |
| Glucoamylase                                  |                            |            |                  |         |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ protein | 63                         | 61         | 5.0              | 0.40    |
| $\mu\text{mol min}^{-1}\text{g}^{-1}$ mucosa  | 9                          | 7          | 0.7              | 0.06    |
| mol/day                                       | 0.8                        | 0.5        | 0.06             | 0.01    |
| Serum xylose (mmol/l)                         | 0.7                        | 0.7        | 0.08             | 0.72    |

<sup>a</sup>Amount of food given per piglet from day 1–5 after weaning: continuous (150, 175, 200, 225 and 250 g/day); restricted (0, 25, 50, 75 and 100 g/day).

<sup>b</sup>SEM: standard error of the mean.

maintenance, 2.5 times maintenance and ad libitum intake) every two hours for five days, there was a linear relationship between total dry matter intake and mean villous height along the length of the small intestine (Fig. 6a). In turn, the mean villous height explained 47% of the total variation in empty body-weight gain in the first five days after weaning (Fig. 6b), a result also found by Li et al. (1991b). These data highlight the interdependence between absorbed nutrients, intestinal structure and growth rate in the immediate post-weaning period, and suggest that if pigs are offered milk liquid diets at regular intervals following weaning, then the growth check could be overcome.

In contrast, pigs consuming a pelleted starter diet at the same level of energy intake as pigs consuming cows' whole milk at 2.5 maintenance (M) showed villous atrophy and crypt hyperplasia, although linear relationships similar to those described above were

also evident (Pluske et al., 1996c). This finding shows an independent effect of diet per se on gut structure after weaning that is uncomplicated by any differences in the level of voluntary energy intake. Despite these marked histological changes between pigs fed milk and solid diets, and the fact that pigs consuming the solid diet grew at similar rates to pigs drinking milk at 2.5 M, we were unable to confirm any differences in gut function as assessed by brush-border enzyme activity and xylose.

The findings of Pluske et al. (1996c) are in apparent contrast to those reported by Beers-Schreurs et al. (1995), who found that pigs fed a commercial diet and sows' milk at the same level of energy intake had similar villous heights and crypt depths. As

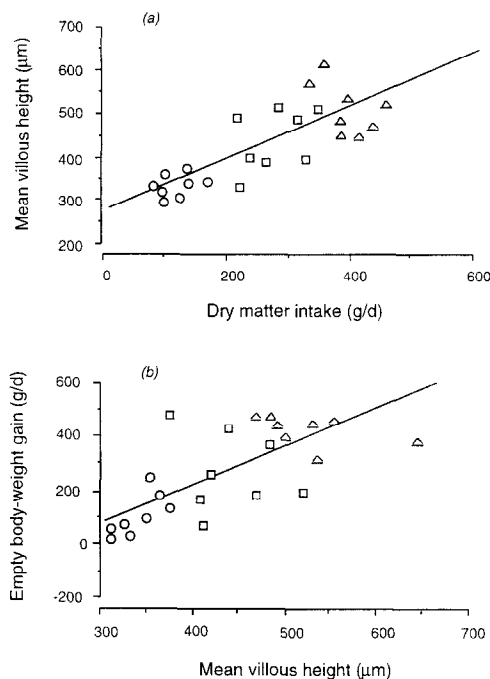


Fig. 6. Relationship between (a) daily dry matter intake and mean villous height along the small intestine [ $y = 286.10 + 0.54x$ ,  $R^2 = 0.68$  (RSD = 56.91);  $P < 0.001$ ], and (b) mean villous height along the small intestine and daily empty body-weight gain [ $y = -325.69 + 1.39x$ ,  $R^2 = 0.48$  (RSD = 127.10);  $P = 0.002$ ], in pigs offered cows' liquid milk at maintenance (○), 2.5 times maintenance (□), or ad libitum (△) energy intake for five days after weaning (8 pigs per treatment). Each point represents a single piglet killed on the fifth day following weaning at 28 days of age (redrawn from Pluske et al., 1995).

reported by Beers-Schreurs (1996), however, the average energy intake of pigs fed these two diets was approximately 2.8 and 2.5 MJ DE/day, respectively, which was about half the daily energy intake achieved by the pigs in the study of Pluske et al. (1996c). This difference most likely explains the differences observed between the two studies.

This discussion has focused primarily on the direct effects of a reduction in energy intake on gut structure and function. It is possible, however, that effects other than a dietary shortage of energy and (or) protein may also contribute to gut morphometry after weaning. In this regard, Williams et al. (1996) found in weanling rats, for example, that feeding a riboflavin-deficient diet for eight weeks from weaning caused a significantly lower villous number, a significant increase in villous length, and an increased rate of enterocyte migration along the villus, in comparison to weight-matched controls. These changes may explain the decreased rate of Fe absorption seen under conditions of riboflavin deficiency.

### 3.5. Cytokines as regulators of epithelial cell growth

Cytokines mediate a variety of important functions within the animal which can be grouped into the general areas of homeopoiesis, innate immunity, and acquired immunity. Cytokines are clearly involved in the communication among lymphoid cells of the mucosal immune system that include Peyer's Patches, lamina propria lymphocytes, and intra-epithelial lymphocytes. A large body of evidence has now amassed, and will continue to do so, concerning the role of cytokines as important regulators of intestinal immunity (see Elson and Beagley, 1994, Gaskins and Kelley, 1995, and Kramer et al., 1995, for recent reviews).

Of particular interest is the role of cytokines as major regulators of epithelial cell growth and development, including intestinal inflammation and epithelial restitution following mucosal damage (Elson and Beagley, 1994). For example, Cunha Ferreira et al. (1990) showed that activation of T cells in the lamina propria of explants of fetal human small intestine caused villous atrophy and crypt hyperplasia, in the absence of damage to surface enterocytes. Similarly, Lionetti et al. (1993), using a similar model of fetal human small intestinal explants,

demonstrated that large numbers of activated macrophages can result in villous atrophy, crypt hyperplasia and, in some cases, complete mucosal destruction.

Cytokine production by epithelial cells almost certainly impacts on the mucosal immune system, and vice versa. The role that such cytokine 'cross talk' between epithelial and lymphoid cells plays in either epithelial integrity or mucosal immune function is not yet fully understood. The role of cytokines in the histological, biochemical and immunological changes that occur in the small intestine of the young pig after weaning has not been explored, and is clearly an exciting area of research in the future.

## 4. Conclusions

In this review, we have described changes to the structure and function of the small intestine in the weaned pig, and outlined factors contributing to these changes. Although we have described each of these factors separately, it should be understood that many of these factors interact with each other, often in a subtle manner, to affect the alterations seen in both mucosal architecture and biochemistry. Since there is an abrupt change of nutrition associated with the weaning process, it is likely that interactions between dietary growth, 'protective' factors and pathogenic microorganisms with the cell epithelium are important determinants of the way the weaned pig digests and absorbs the food it consumes. Furthermore, advances in our understanding in the important area of immunobiology, and interactions between cytokines and the mucosal immune system, will provide greater insights into the mechanisms controlling these functions. Other exciting possibilities for future research in this field include the manipulation of the developing gut with exogenous growth factors, and the use of 'non-essential' amino acids such as glutamine, to maintain the digestive, absorptive and immunological capacity of the small intestine at a time when it is compromised. Finally, we suggest, from an experimental point of view, that future studies in this field consider the profound influence that luminal nutrition has in determining intestinal morphology and function, since results of some previous studies in this field have most likely been confounded by this interaction.



## Acknowledgements

Financial assistance from the Pig Research and Development Corporation (PRDC) of Australia for some of the studies quoted in this review is acknowledged. One of us (JRP) was supported by a Junior Research Fellowship from the PRDC during the course of these experiments. We also acknowledge the invaluable comments and suggestions of an anonymous referee.

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